

What Counts as an Establishment? Measuring Definitional Fragmentation Across Saudi Arabian Legislation

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Abstract

Statutes define their terms, and different statutes define the same terms differently. Whether that matters — whether it represents genuine inconsistency or merely the ordinary business of legislative drafting — has been argued from examples rather than measured. Using a machine-readable corpus of 290 Saudi Arabian legislative instruments carrying 1,920 defined terms and 3,318 definitions, we measure it. The measurement requires two filters that the question’s framing tends to hide. The first separates *indexical* definitions, in which an instrument defines “the Law” as itself or “the Ministry” as its own supervising body: these are the most frequently repeated definitions in the corpus, they are instrument-local by construction, and they cannot disagree. They account for 171 of the 1,920 terms and dominate any raw ranking, the term “the Law” being defined 125 times. The second separates lexical from substantive divergence. Among the 322 substantive terms defined by more than one instrument, 61.5% carry definitions that diverge lexically — a figure that would make a striking headline and that we argue should not be read as legal inconsistency, since the most widely shared term of all, “person”, is defined by 27 instruments in wording that varies while the meaning does not. Reading the twelve most widely shared substantive terms across every instrument that defines them, we find three harmonized, two with instrument-local referents, two indexical cases the automatic filter missed, one instance of homonymy, and four carrying materially conflicting scope — so that within this sample lexical divergence overstates substantive conflict roughly threefold. The clearest is *al-munsha’ah*, “establishment”: under the Competition Law any person carrying on an economic activity, under the Labour Law any undertaking employing at least one worker for wages. A sole trader with no employees is an establishment for one statute and not for the other. We conclude that definitional fragmentation in Saudi legislation is real but far narrower than automatic measurement suggests, that the distinction between lexical and substantive divergence is the methodological crux for any jurisdiction, and that the small set of genuinely conflicting terms is a tractable target for legislative harmonisation and a necessary input for legal AI systems that resolve statutory terms across instruments.

Keywords: Statutory definitions · Legislative drafting · Saudi Arabia · Legal informatics · Terminological consistency · Corpus methods

1 Introduction

Legislative drafting manuals agree that a term should mean the same thing throughout a legal system, and legislative practice does not comply. Statutes define their own terms, definitions accumulate, and the same word ends up carrying different content in different instruments. Practitioners encounter this constantly; the literature discusses it through well-chosen examples. What has been missing is measurement, and the reason is that measurement requires a machine-readable corpus of statutory definitions, which few jurisdictions have.

This paper measures definitional fragmentation across the legislation of Saudi Arabia, using a corpus of 290 instruments in which every definitions article has been parsed into term–definition pairs: 1,920 distinct terms carrying 3,318 definitions (Almohammed, 2026). The corpus makes the question tractable for the first time in an Arab jurisdiction, and Saudi Arabia is a useful case: its legislation was substantially rewritten in the 2020s, so definitions drafted decades apart now sit alongside definitions drafted last year.

The central finding is negative in a way we think is more useful than a striking positive would have been. Measured automatically, definitional fragmentation looks severe: among substantive terms shared by more than one instrument, 61.5% carry lexically divergent definitions. Read by hand, most of that divergence is not legal inconsistency at all. It is variation in wording around a stable meaning, or homonymy, or a referent that each instrument fixes for itself. The residue that does represent conflict is small, specific, and — because it is small and specific — actionable.

Getting from the first number to the second requires two separations that this paper treats as its methodological contribution.

The first is *indexical* versus *substantive* definitions. Saudi instruments open by defining “the Law” as themselves, “the Regulation” as their own implementing regulation, “the Minister” as whichever minister supervises them. These are the most repeated definitions in the corpus — “the Law” appears as a defined term in 125 instruments — and they are instrument-local by construction. Two instruments defining “the Minister” differently are not in conflict; they have different ministers. Any measure that counts them produces a number about drafting convention rather than about legal coherence.

The second is *lexical* versus *substantive* divergence. Automatic similarity measures compare wording. Legal consistency is a property of meaning. The term “person” is defined by 27 Saudi instruments, in formulations ranging from “the natural or legal person” to “any person of natural or legal capacity, public or private”; lexical overlap between some pairs is near zero and the legal content is identical. Reporting the lexical figure as fragmentation would be, in a precise sense, measuring the wrong thing.

Section 3 sets out both separations and the hand-adjudication procedure. Section 4 reports what survives them. Section 5 argues that the surviving cases matter for legislative harmonisation and for legal AI, and Section 6 sets out what the method cannot see.

2 Related work

Statutory definitions. Definitions are a recognised topic in legislative drafting scholarship, where the standard guidance is that a defined term should be used consistently and that definitions should

not vary between related instruments without reason. The scholarship is largely doctrinal and example-driven; corpus-scale measurement of whether a legal system actually achieves definitional consistency is rare, and we are aware of none for an Arab jurisdiction.

Legal informatics and legal corpora. Machine-readable legal corpora have grown substantially — the Pile of Law (Henderson et al., 2022), MultiLegalPile (Niklaus et al., 2024) — but they are built for language-model training rather than for structural analysis, and none covers Saudi legislation. Structural standards such as Akoma Ntoso (Palmirani and Vitali, 2011) model legislative documents without extracting definitional content. The corpus used here (Almohammed, 2026) ships a glossary layer that parses definitions articles into term–definition pairs with provenance, which is what makes this study possible.

Quantitative studies of legal systems. Work measuring legal complexity and structure at corpus scale (Katz and Bommarito, 2014; Coupette et al., 2021) has concentrated on citation and on document structure rather than on terminology. A companion study analyses the citation network of the same corpus; the present paper takes the complementary question of whether the instruments in that network speak a common language.

3 Data and method

3.1 Corpus and glossary

Saudi statutes conventionally open with a definitions article of the form “the following words and phrases, wherever they appear in this Law, shall have the meanings assigned to them”. A parser over those articles yields 1,920 distinct terms carrying 3,318 definitions, drawn from the 214 instruments whose definitions articles parse cleanly (Almohammed, 2026). Each definition record carries the term as written, the definition text, the instrument, and the article it came from.

3.2 Filter one: indexical definitions

A definition is *indexical* when it names a particular instrument or organ of state rather than describing a class: “the Law” defined as “the Agriculture Law”, “the Ministry” as “the Ministry of Environment, Water and Agriculture”. Such definitions are local to the enacting instrument by design and cannot be inconsistent with one another.

We detect them by a rule that is deliberately simple and inspectable. A definition counts as indexical when it is at most twelve content tokens long, is headed by an instrument or state-organ word (law, regulation, ministry, minister, authority, council, court, body, and so on), and does *not* open with a genus word. The last condition carries most of the weight: Arabic statutory drafting puts the genus first, so a class definition begins “any natural person who...” or “a document issued by...”, while a naming definition does not. A term is treated as indexical when the majority of its definitions are.

Two implementation details proved to matter more than the rule itself, and we record them because they are the kind of thing that silently corrupts this sort of study. First, Arabic normalisation must be applied to the keyword lists as well as to the text: comparing unnormalised keywords

Measure	Value
Distinct defined terms	1,920
Definition occurrences	3,318
Instruments with a parsed definitions article	214
Indexical terms	171
Substantive terms	1,749
Terms defined by more than one instrument	362
of which indexical	40
of which substantive	322
Substantive shared terms, lexically identical	94
Substantive shared terms, near-identical	30
Substantive shared terms, lexically divergent	198 (61.5%)

Table 1: Definitional structure of the corpus.

against normalised tokens matches nothing for every head ending in *teh marbuta*, which silently reclassified “the Regulation”, “the Ministry” and “the Authority” as substantive. Second, the genus test must be restricted to the head of the definition: the relative pronouns *man* and *ma* are common inside naming definitions and, treated as genus words anywhere in the string, exclude them wrongly.

3.3 Filter two: lexical divergence

For each substantive term defined by more than one instrument, we take one definition per instrument and compute pairwise Jaccard similarity over normalised content tokens. A term is *lexically identical* when the worst pair exceeds 0.95, *near-identical* between 0.75 and 0.95, and *lexically divergent* below 0.75.

We report this measure and then argue against reading it as legal inconsistency. It is included because it is what an automatic pipeline produces, and because the gap between it and the hand-adjudicated result is the paper’s point.

3.4 Filter three: hand adjudication

The twelve most widely shared substantive terms were read across every instrument that defines them and placed in one of five classes: *harmonized* (same concept and scope, varied wording); *instrument-local* (same concept, referent fixed by the enacting instrument); *indexical* (a naming definition the automatic filter missed, reported rather than silently corrected); *homonymous* (different concepts sharing a label); and *conflicting* (same concept, materially different scope). The adjudication table is published with the analysis code so that each assignment can be checked.

Twelve terms is a small sample, chosen because these are the terms shared by the most instruments and therefore the ones where inconsistency would do the most damage. We make no claim about the remaining shared terms beyond the lexical measure.

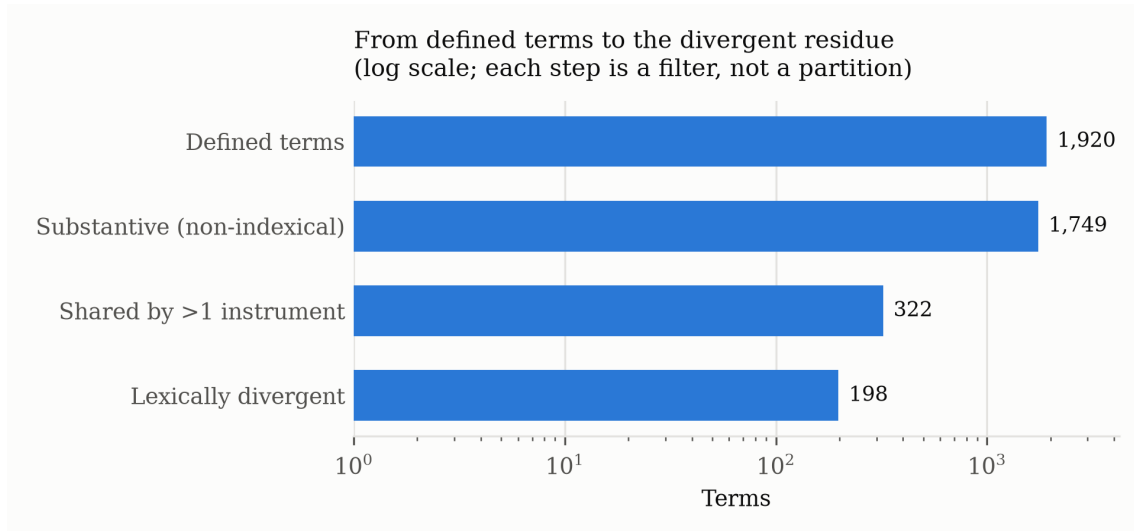


Figure 1: From all defined terms to the lexically divergent residue. Each step is a filter: 1,920 defined terms, 1,749 of them substantive, 322 of those shared by more than one instrument, 198 of which diverge lexically.

4 Results

4.1 The funnel

Table 1 and Figure 1 give the shape of the data. Two features deserve comment before any interpretation.

First, **most defined terms are not shared at all**. Of 1,920 terms, 1,558 are defined by exactly one instrument. Saudi legislative drafting is overwhelmingly local: each statute builds its own vocabulary and only 19% of terms are touched by more than one instrument. Fragmentation, whatever its extent, is confined to a small part of the lexicon.

Second, **indexical definitions dominate the top of any raw ranking**. The most frequently defined terms in the corpus are “the Law” (125 instruments), “the Regulation” (118), “the Minister” (85), “the Ministry” (75), “the Authority” (67) and “the Council” (60). All are indexical. A study that reported these as the most contested terms in Saudi law would be reporting a drafting convention.

4.2 Lexical divergence overstates conflict

Among the 322 substantive shared terms, 198 (61.5%) are lexically divergent. Read by hand, that figure does not survive (Figure 2).

Of the twelve most widely shared substantive terms, three are **harmonized**: “person”, “licensee” and “employer” carry the same legal content wherever they appear, in wording that varies. “Person” is the extreme case — 27 instruments, lexical similarity near zero between some pairs, and no legal difference whatever between “the natural or legal person” and “any person of natural or legal capacity”.

The adjudication also measures the automatic filter against a hand-read sample, which is the only recall figure we can offer for it: of the twelve terms, two turned out to be indexical definitions

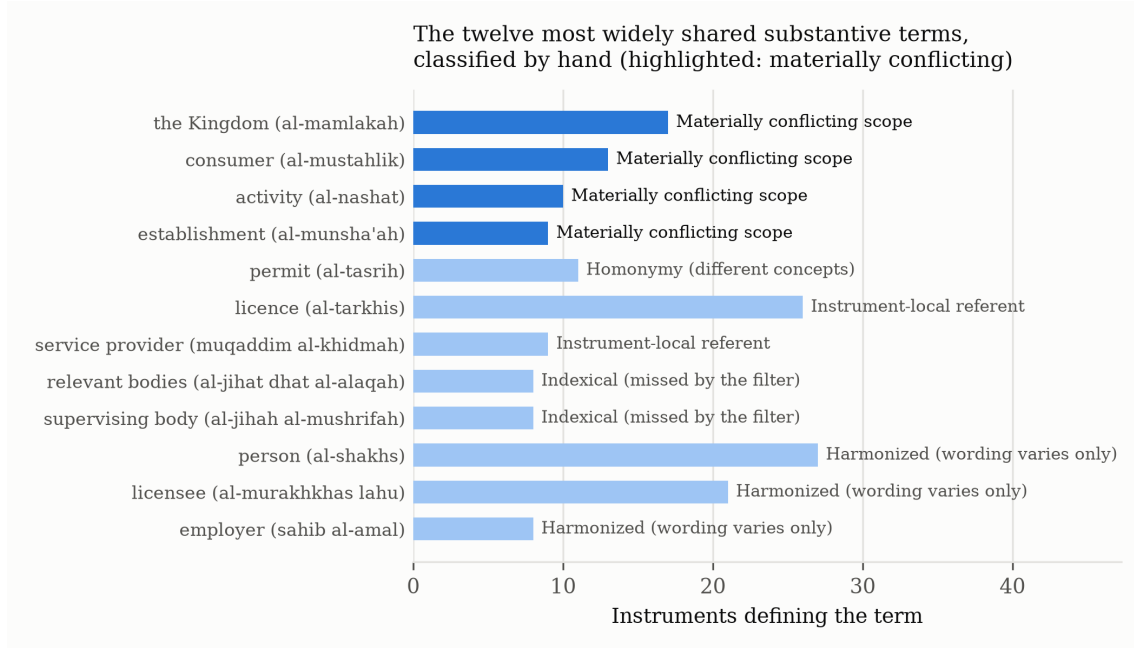


Figure 2: The twelve most widely shared substantive terms, classified by hand across every instrument that defines them. Four carry materially conflicting scope; the rest diverge in wording, referent or concept rather than in law.

the filter classified as substantive, so its recall on widely shared terms is around 83%. We report the misses in place rather than correcting them silently, since correcting them would inflate the filter’s apparent accuracy on the very sample used to assess it.

Two are **instrument-local**: “licence” and “service provider” denote the same kind of thing everywhere, an authorisation and its holder, with each instrument fixing the sector and the issuing body. Two more are **indexical** definitions that the automatic filter did not catch — “relevant bodies” and “supervising body” each name whichever government entities the enacting instrument designates.

One is **homonymy**: “permit” covers a temporary agricultural authorisation in one instrument, an aircraft overflight clearance in another, and a pre-commencement environmental document in a third. These are different concepts wearing the same label, which is a feature of natural language rather than a defect of legislation.

4.3 The four conflicting terms

Four terms carry materially different scope for what is recognisably the same concept.

Al-munsha’ah (“establishment”), defined by nine instruments, is the clearest. The Competition Law and its regulation define it as any natural or legal person carrying on an economic activity. The Anti-Concealment Regulation extends it to “anyone practising an economic activity, including the sole proprietorship, the company, and any other legal form”. The Labour Law defines it instead as any undertaking run by a natural or legal person that employs one or more workers for wages. The difference is not stylistic: a sole trader with no employees is an establishment for competition purposes and is not one for labour purposes. Any system — human or automated — that resolves “establishment” without asking which statute is speaking will be wrong for some population of

businesses.

Al-mustahlik (“consumer”), defined by thirteen instruments, ranges from a person with credit dealings, in the credit-information regime, to a residential or commercial consumer in the electricity regime, to a person transacting electronically in the e-commerce regime. The regimes are adjacent enough that the divergence has practical consequences for who enjoys consumer protections in a transaction that spans them.

Al-mamlakah (“the Kingdom”), defined by seventeen instruments, is usually simply “the Kingdom of Saudi Arabia”. Instruments with an extraterritorial reach define it as the territory of the Kingdom *including* areas beyond the territorial waters over which the Kingdom exercises sovereign rights. For a term that fixes jurisdictional scope, that is a substantial difference, and it is invisible to anyone reading a single instrument.

Al-nashat (“activity”), defined by ten instruments, runs from “a type of subsidiary work within a field” to “any project or industrial, commercial or service facility expected to have environmental effects” to commercial activity carried on for profit. Here the divergence approaches homonymy, and we classify it as conflicting because the instruments concerned regulate overlapping conduct.

5 Discussion

The measurement problem generalises. Any attempt to measure definitional consistency automatically will encounter both filters described here. Indexical definitions exist wherever instruments name themselves and their own institutions, which is to say nearly everywhere; and lexical similarity will always be a poor proxy for legal identity, in any language. A study that reports an unfiltered figure is reporting drafting convention plus paraphrase. The size of the gap is worth stating precisely, because it is easy to overstate: all twelve adjudicated terms are lexically divergent, and four of them are substantively conflicting, so within this sample lexical divergence overstates substantive conflict roughly *threefold*. That ratio is measured on twelve terms and should not be projected onto the other 310 shared substantive terms, nor onto other jurisdictions; measuring it elsewhere would be a useful test of whether a similar factor holds.

Harmonisation is tractable. The practical implication for legislative drafting is optimistic. If definitional conflict in Saudi legislation were pervasive, harmonisation would be a generational project. If it is concentrated in a handful of high-traffic terms — establishment, consumer, the Kingdom, activity — then a drafting body could address it deliberately, and the corpus makes the candidate list computable rather than anecdotal.

Consequences for legal AI. Systems that answer questions over statutory text must resolve terms, and the conflicting cases are exactly where naive resolution fails. A retrieval-augmented system asked whether a sole trader is subject to an obligation imposed on “establishments” will retrieve a definition — and which definition it retrieves determines the answer. Term resolution in legal AI is therefore not a lookup but a choice of governing instrument, and a system that does not model that choice will be confidently wrong on precisely the terms that matter most.

6 Limitations

The glossary is parsed, not curated. Definitions come from an automatic parser over definitions articles. We observed occurrences where one definition’s text bleeds into the next, and terms defined outside the definitions article are not captured at all. The counts here are therefore approximate at the margins.

The indexical filter is a heuristic. It is inspectable and we report its misses — two of the twelve adjudicated terms are indexical definitions it did not catch — but it has no measured precision or recall against a hand-labelled sample. Building that sample is the obvious next step.

The adjudication is single-annotator and small. Twelve terms were classified by one reader. There is no inter-annotator agreement to report, the classes involve judgement at the boundaries (“activity” could be defended as homonymy rather than conflict), and the finding should be read as a documented reading rather than a measurement.

The sample is chosen by frequency, which may understate conflict. The twelve adjudicated terms are those shared by the most instruments, and frequency selects for generality: widely shared terms tend to be broad concepts (“person”, “licence”) that are easy to harmonise or that diverge only by sector. Genuinely conflicting *technical* terms, shared by two or three instruments in adjacent regimes, would sit in the tail and are not examined here. The four-in-twelve figure is therefore a lower bound on the density of conflict in the shared lexicon, not an estimate of it.

Divergence is not the same as inconsistency. Two instruments may define a term differently for good reason, because they regulate different conduct. Nothing here establishes that the four conflicting cases are drafting errors; the claim is that they are places where the same word carries different legal content, which is a fact about the system that its users need to know.

Coverage. Only the 214 instruments whose definitions articles parse cleanly contribute. Instruments without a definitions article, and those the parser skipped, are invisible to the study.

7 Conclusion

Definitional fragmentation in Saudi legislation is real, narrow, and measurable. Of 1,920 defined terms, most are used by a single instrument; of the 322 substantive terms that more than one instrument defines, 61.5% diverge lexically; and of the twelve most widely shared, four carry materially different legal scope. The gap between the second figure and the third is the paper’s main methodological claim: measuring statutory definitional consistency automatically requires separating indexical from substantive definitions and lexical from substantive divergence, and a study that does neither will report a number about drafting convention and paraphrase rather than about law.

The four conflicting terms are, in a sense, the smallest possible finding and the most useful one. *Establishment* means one thing to the competition regime and another to the labour regime,

and a sole trader falls on different sides of that line depending on which statute is asking. That is a fact a practitioner may know and a machine cannot infer — and it is now, for this jurisdiction, enumerable.

Declarations

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Competing interests. The author declares no competing interests.

Ethics approval. Not applicable. The study involves no human participants, no animal subjects, and no personal data; it analyses published national legislation only.

Data availability. The corpus, its glossary layer, and the analysis and figure scripts that reproduce every number and figure reported here are openly available at <https://github.com/Abdullah-m7/saudi-legal-corpus-ai> and archived on Zenodo under the MIT licence (10.5281/zenodo.22019183; concept DOI 10.5281/zenodo.22019182).

Author contributions. A. Almohammed is the sole author and conducted all aspects of the work.

Legal status. This paper analyses a non-official research corpus. Nothing here constitutes legal advice; the binding text of any instrument is the Arabic original published in *Umm Al-Qura*.

References

Abdullah Almohammed. The Saudi Legal Corpus for AI: An auditable, official-source-grounded, article-level corpus of Saudi Arabian legislation, 2026. Manuscript under review. Corpus archived on Zenodo, DOI 10.5281/zenodo.22019183.

Corinna Coupette, Janis Beckedorf, Dirk Hartung, Michael Bommarito, and Daniel Martin Katz. Measuring law over time: A network analytical framework with an application to statutes and regulations in the United States and Germany. *Frontiers in Physics*, 9, 2021.

Peter Henderson, Mark S. Krass, Lucia Zheng, Neel Guha, Christopher D. Manning, Dan Jurafsky, and Daniel E. Ho. Pile of Law: Learning responsible data filtering from the law and a 256GB open-source legal dataset. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022), Datasets and Benchmarks Track*, 2022.

Daniel Martin Katz and Michael J. Bommarito. Measuring the complexity of the law: The United States Code. *Artificial Intelligence and Law*, 22(4):337–374, 2014.

Joel Niklaus, Veton Matoshi, Matthias Stürmer, Ilias Chalkidis, and Daniel E. Ho. MultiLegalPile: A 689GB multilingual legal corpus. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, Bangkok, Thailand, 2024. Association for Computational Linguistics.

Monica Palmirani and Fabio Vitali. Akoma-Ntoso for legal documents. In *Legislative XML for the Semantic Web*, pages 75–100. Springer, 2011.